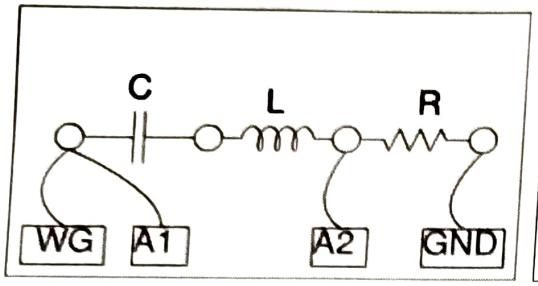
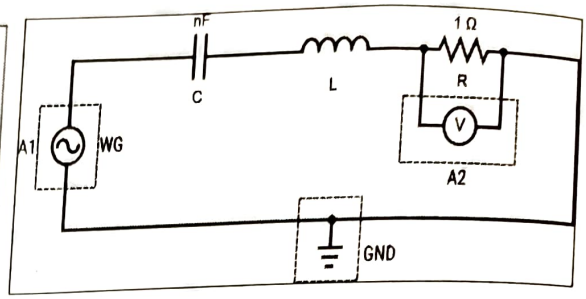
SERIES L-C-R CIRCUIT

OBSERVATIONS:

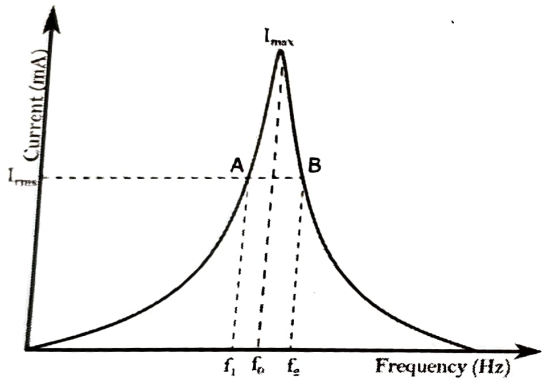
ExpEyes circuit:



Conventional Circuit:



Model Graph:



Formulae

Capacitance C = nF

$$L = \frac{1}{4\pi^2 f_0^2 C}$$ = H

I_{max} = mA

$$I_{RMS} = \frac{I_{max}}{\sqrt{2}}$$ = mA

Resonant Frequency f_0 = Hz

Lower Cut off frequency f_1 = Hz

Upper Cut off frequency f_2 = Hz

Band width $\Delta f = f_2 - f_1$ = Hz

Quality Factor $Q = \frac{f_0}{\Delta f}$ =

Calculations:

SERIES L-C-R CIRCUIT

Experiment No:

Date:

Aim: To study the frequency response of LCR circuit and determination of a) Self-inductance of the given coil, b) Quality factor (Q- value) and c) Band-width

Apparatus: expEYES-17 kit, Resistance box, Capacitor, Inductor

Principle: The phenomenon of resonance is common among systems that have a tendency to oscillate at a particular frequency. This frequency is called the system's natural frequency. If such a system is driven by an energy source at a frequency that is near the natural frequency, the amplitude of oscillation is found to be large. An interesting characteristic of the series RLC circuit is the phenomenon of resonance. Resonance occurs when the inductive reactance equals the capacitive reactance, i.e., $X_L = X_C$. $\omega L = \frac{1}{\omega C}$; At resonance, the impedance of the circuit is purely resistive, and the circuit's total impedance is minimized, leading to maximum current flow.

The resonant angular frequency ω_0 is: $\omega_0 = \frac{1}{\sqrt{LC}}$

The corresponding resonant frequency f_0 is: $f_0 = \frac{1}{2\pi\sqrt{LC}}$

At Resonance:

- The voltage across the inductor and capacitor cancel each other out, leaving the voltage across the resistor to equal the applied voltage.
- The circuit exhibits a maximum current I_{max} given by: $I_{max} = \frac{V}{R}$, where V is the applied voltage.

Theory:

A Series L-C-R circuit is a fundamental electrical circuit consisting of three components: an inductor (L), a capacitor (C) and a resistor (R), all connected in series. The principle of the Series L-C-R circuit lies in the interplay between the inductor, capacitor, and resistor when an AC voltage is applied across the circuit. The inductor and capacitor store energy in magnetic and electric fields, respectively and exhibit opposition to changes in current and voltage while the resistor dissipates energy as heat.

Inductor (L): An inductor opposes varying current by generating a back EMF (electromotive force). The voltage across the inductor leads the current by 90°.

Capacitor (C): A capacitor opposes changes in voltage by storing energy in an electric field. The current through the capacitor leads the voltage across it by 90°.

Resistor (R): A resistor opposes current by dissipating energy as heat, with no phase difference between the voltage across and the current through it.

Impedance (Z) of the Circuit:

- In a Series L-C-R circuit, the total impedance (Z) is the vector sum of the individual impedances of the resistor, inductor, and capacitor.
- *Resistive Impedance (R):* The impedance due to the resistor is purely real and is equal to the resistance *R*.

Note: Graph plot procedure using GNUPLOT

Go to respective folder/directory on the desktop.

Right click on the mouse, **select open in terminal**

type `gnuplot` **in the terminal and press ↵ Enter**

Sample code

```
set timestamp
set title 'Series LCR circuit-Rajesh'
set xlabel 'Frequency in Hz'
set ylabel 'Current in mA'
plot 'lcr.txt' w lp
```

Marking points on the graph

- Take the cursor to the tip of the curve
- Press the scroll button on the mouse
- Note down the corresponding frequency (x co-ordinate) as resonant frequency f_0 and the corresponding current(y co-ordinate) as maximum or resonant current I_{max}
- Find out I_{RMS} $(=\frac{I_{max}}{\sqrt{2}})$
- Press the scroll button on the mouse at I_{RMS} on the left side of the graph and note down the corresponding frequency as lower cut off frequency f_1.
- Press the scroll button on the mouse at I_{RMS} on the right side of the graph and note down the corresponding frequency as upper cut off frequency f_2.
- Export the file as PDF

Result:

1.	Resonant frequency of the circuit f_0	Hz
2.	Self-inductance of the given coil L	H
3.	Quality factor from graph Q	
4.	Band width $\Delta f = f_2 - f_1$	Hz

Inductive reactance (X_L): The reactance due to the inductor is given by: $X_L = \omega L$ where $\omega = 2\pi f$, ω is the angular frequency and f is the frequency of the applied AC signal.

Capacitive reactance (X_C): The reactance due to the capacitor is given by: $X_C = \frac{1}{\omega C}$

Total Impedance (Z): The total impedance in the series L-C-R circuit is:
$|Z| = \sqrt{R^2 + (X_L - X_C)^2}$

Quality Factor (Q-Factor):

- The Quality Factor (Q-factor) is a measure of the sharpness of resonance in the Series L-C-R circuit. It is given by: $Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR}$, A higher Q-factor indicates a sharper resonance peak, meaning the circuit is more selective in its response to frequencies near the resonant frequency.

Power in Series L-C-R Circuit:

- Power Dissipation: Power is dissipated only in the resistor and is given by:

$$P = I^2 R = \frac{V^2 R}{Z^2}$$

Applications:

- Tuning Circuits, Filters, Impedance Matching etc.,

Procedure:

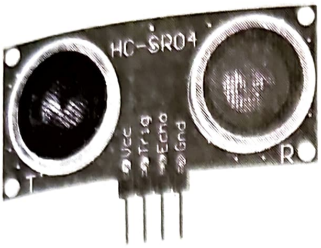
1. Make the connections as per the circuit diagram.
2. Select the frequency range in the right side of GUI window.
3. Enter the value of resistance in the box (say 1Ω).
4. Enter the frequency interval steps in the GUI window.
5. Click on START button to plot the current versus frequency.
6. Save the data by clicking on the Sava Data button.
7. A graph of frequency versus current is plotted using GNUPLOT software, this graph is called frequency response of the L C R circuit.
8. Note down the corresponding frequency (x co-ordinate) as resonant frequency f_0 and the corresponding current(y co-ordinate) as maximum or resonant current I_{max} Find out I_{RMS} $(=\frac{I_{max}}{\sqrt{2}})$.
9. Press the scroll button on the mouse at I_{RMS} on the left side of the graph and note down the corresponding frequency as lower cut off frequency f_1.
10. Press the scroll button on the mouse at I_{RMS} on the right side of the graph and note down the corresponding frequency as upper cut off frequency f_2.
11. Find the self-inductance of the given coil, Quality factor and Band width Δf using the relevant formulae.

Results:

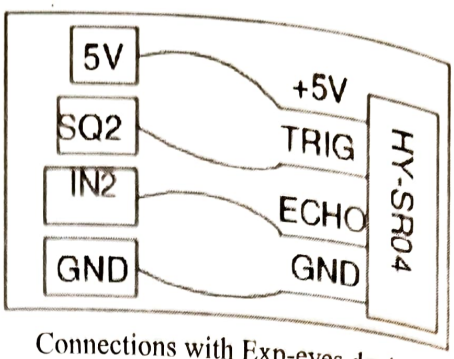
1.	Resonant frequency of the circuit f_0	Hz
2.	Self-inductance of the given coil L	H
3.	Quality factor from graph Q	
4.	Band width $\Delta f = f_2 - f_1$	Hz

SPRING CONSTANT

OBSERVATIONS

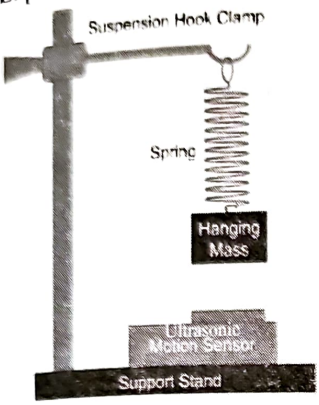


Ultrasonic sensor (HC-SR04)

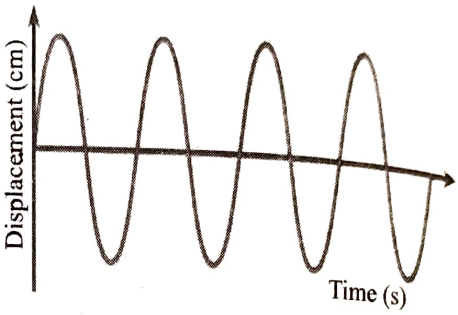


Connections with Exp-eyes device

Experimental setup



Model graph



Formula: Spring constant is given by $K = 4\pi^2 \frac{m}{T^2}$ N/m

Where, m is the mass of the load (kg). T is the time period (s) and K is the spring constant N/m.

Graph plotting with GNUPLOT

Go to respective folder in which the file is saved.

Right click at the empty area, select **'open in terminal'** and Run the following commands:

```
gnuplot
set timestamp
set title 'Spring Constant-Raj'
set xlabel 'Time in S'
set ylabel 'Displacement in cm'
set multiplot layout 2,2 columns
plot 'spring1.txt' w lp
plot 'spring2.txt' w lp
plot 'springseries.txt' w lp
plot 'springparallel.txt' w lp
```

SPRING CONSTANT USING ULTRASONIC SENSOR

Experiment No: 8

Date: 19-12-2024

Aim: Determine spring constant for the given spring using ultrasonic sensor.

Apparatus: springs, weight hanger, slotted weights, stop watch.

Principle: Spring constant (or force constant) of a spring is given by $K = \frac{\text{Restoring Force}}{\text{Extension}}$ N/m

Spring constant is the restoring force per unit extension in the spring. Its value is determined by the elastic properties of the spring. When material is subjected to strain, stress is produced. The restoring force is always directed opposite to the displacement. When a mass m suspended by means of spring is displaced through a small distance and then released, it undergoes simple harmonic motion.

The time period **T** of oscillations of a spring is given by the relation,

$$T = 2\pi \sqrt{\frac{m}{K}} \text{ s}$$

By finding the time period (T) spring constant K can be determined.

Procedure:

1. Suspend one end of the given spring to a rigid support and add the slotted weights at the other end. And note down the mass m of added slotted weights and weight hanger (m_1).
2. Below the load, place an ultrasonic sensor (HC-SR04) connected to the expEYES-17 hardware.
3. Keep the distance of about 6cm between the ultrasonic sensor and the load.
4. Interface the experimental circuit with the CPU and click on mechanics tab and select distance by HC-SR04 echo module experiment.
5. On the screen enter the y axis value as 10cm and time as 10 seconds.
6. Pull the load slightly downwards and then release it gently so that it is set into oscillations in a vertical plane about its mean position.
7. Click on the start button and see the graph of the wave motion generated on the screen.
8. Click on the 'fit curve using sine' button to generate a sine curve on the existing graph.
9. Note down the frequency in Hz and find time period of the oscillation. And then compute the spring constant value (K_1) for the load (m_1).
10. Repeat the same procedure for another identical spring (K_2) for the load (m_2)
11. Connect two springs in series combination and repeat the above activity with an attached mass m_s and find the spring constant for the series combination K_S. Calculate the effective spring constant K_{eff} for series combination. compute the error $K_S - K_{eff}$.
12. Connect two springs in parallel combination and repeat the above activity and find the spring constant K_P for the parallel combination. Calculate K_{eff} for parallel and compute the error $K_P - K_{eff}$

SPRING CONSTANT

OBSERVATIONS

Table 1: Spring constant K_1 for the given material
(Mass of the hanger + Mass of the slotted weights) = m_1 = 0.15 Kg

Trial No	Frequency(Hz)	Time for one osc T_1 (s)	T_1^2	Spring constant $K_1 = 4\pi^2 \frac{m_1}{T_1^2}$ (N/m)
1	1.12 Hz	0.89s	0.7921	7.4684

Table2: Spring constant K_2 for the given material
(Mass of the hanger + Mass of the slotted weights) = m_2 = 0.15 Kg

Trial No	Frequency(Hz)	Time for one osc T_2 (s)	T_2^2	Spring constant $K_2 = 4\pi^2 \frac{m_2}{T_2^2}$ (N/m)
1	1.12 Hz	0.90s	0.81	7.3034

Table 3: Spring constant K_S in series combination for the given material
(Mass of the hanger + Mass of the slotted weights) = m_s = 0.15 Kg

Trial No	Frequency(Hz)	Time for one osc T_S (s)	T_S^2	Spring constant $K_S = 4\pi^2 \frac{m_s}{T_s^2}$ (N/m)	$K_{eff} = \frac{K_1 K_2}{K_1 + K_2}$ (N/m)	Error = $K_{eff} \sim K_S$ (N/m)
1	0.79Hz	1.27s	1.6129	3.1677	3.8924	0.02478

Table 4: Spring constant K_P in parallel combination for the given material
(Mass of the Scale + Mass of the hanger + Mass of the slotted weights) = m_P = 0.1668 Kg

Trial No	Frequency(Hz)	Time for one osc T_P (s)	T_P^2	Spring constant $K_P = 4\pi^2 \frac{m_p}{T_P^2}$ (N/m)	$Kp = K_1 + K_2$ (N/m)	Error= $K_{eff} \sim K_P$ (N/m)
1	1.45Hz	0.69s	0.4761	13.8171	14.7718	0.9547

Results:

a)	The spring constant for the given springs are	K_1 = 7.4684 N/m K_2 = 7.3034 N/m
b)	The spring constant in series combination is	Ks = 3.6677 N/m
c)	The spring constant in parallel combination is	Kp = 13.8171 N/m

About hardware and software:

The circuit consists of a microcontroller system, an ultrasonic sensor, Expeyes hardware and a spring mass system. A microcontroller is a compact integrated circuit designed to govern a specific operation in an embedded system. An ultrasonic sensor measures the distance of an object using sound waves. The ultrasonic sensor consists of 4 pins (Voltage pin, Trigger Pin, ECHO pin, GND pin). Connections are made as shown in the figure, voltage pin is connected to +5V of the expEYES-17; the trig pin of the sensor is connected to the SQ2 of the expEYES-17; the echo pin is connected to the IN2; GND pin is connected to the GND of the expeyes-17. The expeyes-17 interfaced with CPU of the computer. It combines the real-time measurement capability of micro-controllers with the ease and flexibility of Python programming language for data analysis and visualization.

CALCULATIONS:

$\Rightarrow T_1 = 0.895$, $T_1^2 = 0.7921$

$K_1 = 4\pi^2 \frac{m_1}{T_1^2} = 4 \times (3.14)^2 \times \frac{0.15}{0.7921} = 7.4884$ N/m

$\Rightarrow T_2 = 0.905$, $T_2^2 = 0.81$

$K_2 = 4\pi^2 \frac{m_2}{T^2} = 4 \times (3.14)^2 \times \frac{0.15}{0.81} = 7.3034$ N/m

$\Rightarrow T_s = 1.27$, $T_s^2 = 1.6129$

$K_s = 4\pi^2 \frac{m_s}{T_s^2} = 4 \times (3.14)^2 \times \frac{0.15}{1.6129} = 3.6677$ N/m

$K_{eff_s} = \frac{K_1 K_2}{K_1 + K_2} = \frac{7.48 * 7.30}{7.48 + 7.30} = 3.6924$, $K_{eff_s} - K_s = 0.02478$

$\Rightarrow T_p = 0.695$, $T_p^2 = 0.4761$

$K_p = 4\pi^2 \frac{m_p}{T_p^2} = 4 \times (3.14)^2 \times \frac{0.1618}{0.4761} = 13.8171$

$K_{eff_p} = K_1 + K_2 = 7.48 + 7.30 = 14.7781$

$K_{eff_p} - K_p = 0.9547$

29/30 5.26/12